EXPERIMENT 5: SEMANTIC NETWORK USING NETWORKX

Aim:

To represent concepts and labeled relationships as a directed semantic network.

Software Required: Python 3.x and NetworkX.

Install NetworkX using:

pip install networkx

Procedure / Algorithm:

1. Create a directed graph.
2. Add concepts as nodes through edges.
3. Add labeled relationships between concepts.
4. Display each relationship.

Program:

```
G = [
   ("Dog", "is-a", "Mammal"),
   ("Mammal", "is-a", "Animal"),
   ("Dog", "has-instance", "Tommy")
]


print("Semantic Network Relationships:")


for source, relation, target in G:
   print(source, "--", relation, "-->", target)
```

Sample Input:

The concepts and relationships are predefined in the program.

Expected Output:

```
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/saimr/AppData/Local/Programs/Python/Python312/KE-5.py
Semantic Network Relationships:
Dog -- is-a --> Mammal
Mammal -- is-a --> Animal
Dog -- has-instance --> Tommy
>>>
```

Result:

Thus, the semantic network represented concepts as nodes and their relationships as labeled edges.